

Execution-Aware A* Search for Cross-Exchange Stablecoin Arbitrage

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https://github.com/kevin103/Stablecoin-CrossExchange-Arbitrage

Motivation

Prior arbitrage work stresses *detecting* negative cycles, not *executing* trades under fees, slippage, chain latency, and venue risk. Stablecoins (> \$300B market cap) are ideal for cross-exchange routing because USD-pegged balances define a consistent notion of profit across venues.

Contribution

- Cast stablecoin arbitrage as **A*** pathfinding on a **weighted directed graph** whose edges bundle realistic execution rates (trades + transfers).
- Three **domain guidance** penalties h_1 (liquidity), h_2 (order-book slippage), h_3 (congestion/reliability) in $f = g + h$ (penalties, not admissible lower bounds), plus a parallel multi-start baseline.
- Empirical study on a **CEX stablecoin dataset** (12 exchanges, live CCXT feeds), with cached graphs, scaling, an 8-hour campaign, and quote-staleness checks.

Graph model

Nodes: (**exchange**, **stablecoin**). Edges: **intra-exchange trades** and **cross-exchange transfers** of the *same* symbol. Weight $w(e) = -\log r(e)$ for effective rate $r(e)$ (fees, slippage, gas, delay, risk baked in). **Open goal**: any node with final USD value > start; **no** requirement to return to the start token.

Heuristics (summary)

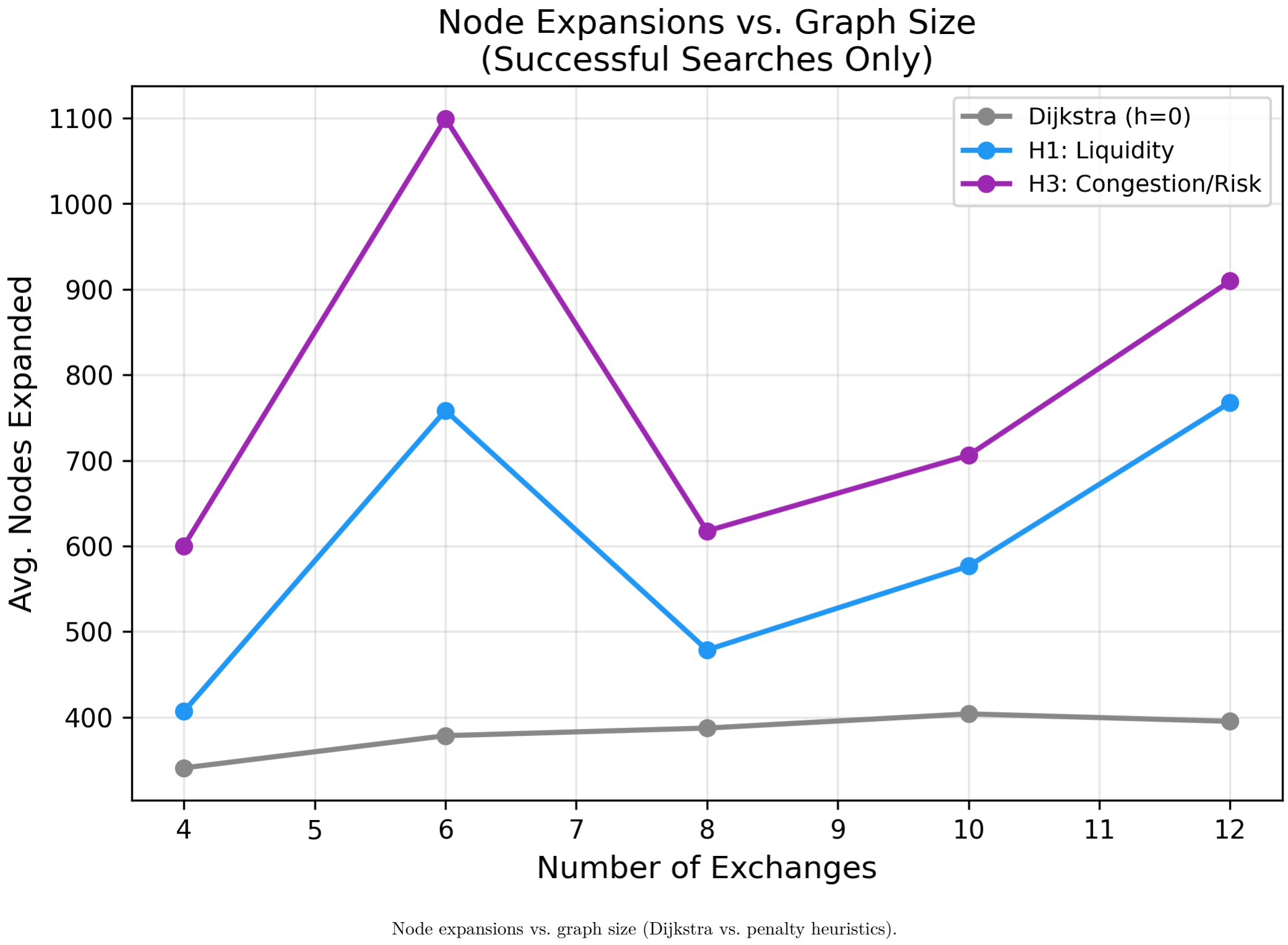
- h_1 : penalize illiquid books (24h volume vs. order size and time left).
- h_2 : **slippage** from live depth / VWAP vs. mid; key result below.
- h_3 : chain time + static exchange reliability; uses weighted A* where needed.

Slippage: $h_2 = \lambda_{\text{OB}} \max\{0, \text{slippage}_{\text{open}} - \theta\}$.

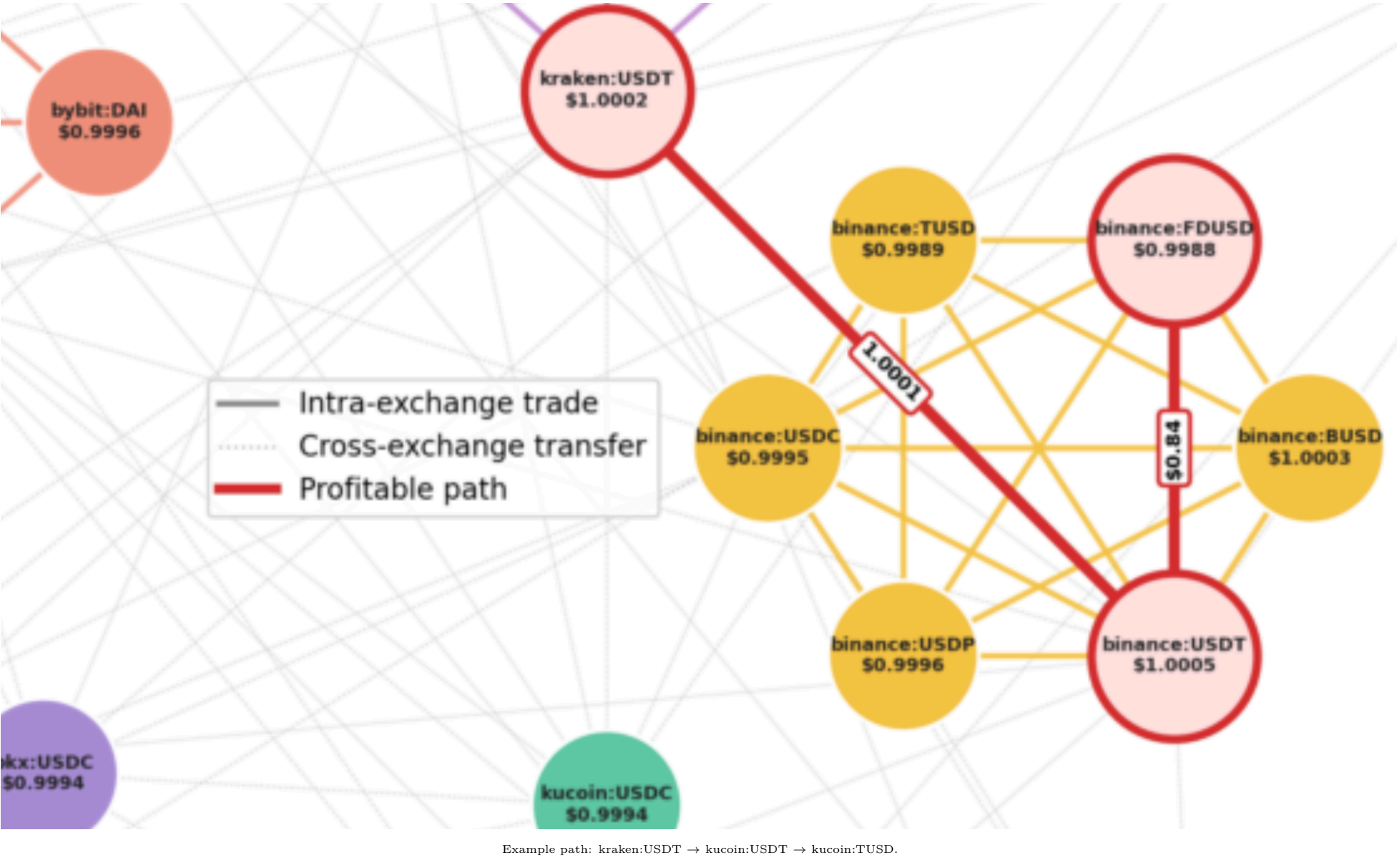
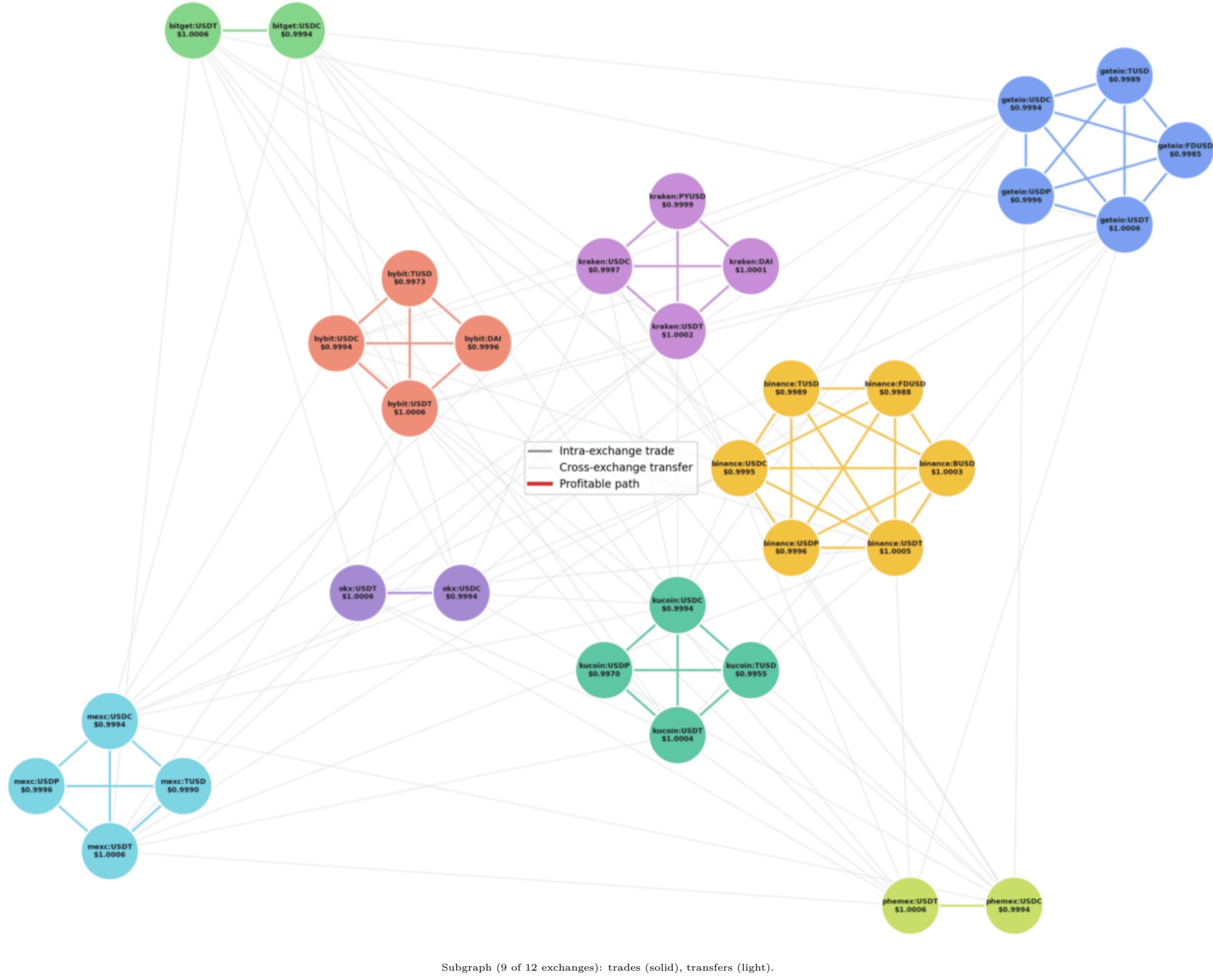
Key numbers (cached graph, \$10k)

Method	Succ.	Profit	Exp.
Dijkstra / $h=0$	56.7%	\$10.01	359
h_2 (slippage)	56.7%	\$9.91	256
h_1	56.7%	\$6.72	598
h_3	56.7%	\$7.06	673

h_2 expands **29% fewer** nodes than Dijkstra with profit within 1%. Overnight campaign (**7,200** instances): every A*-based run finds a profitable path; staleness: **99.6%** still profitable up to 120s under delayed re-quotes.



Illustration



Takeaway

h_2 (**slippage**) best tracks optimal routes while pruning search. Liquidity and congestion penalties can *over-steer* and hurt profit on the cached graph. Future: async order-book prefetch; DEX/AMM extensions.